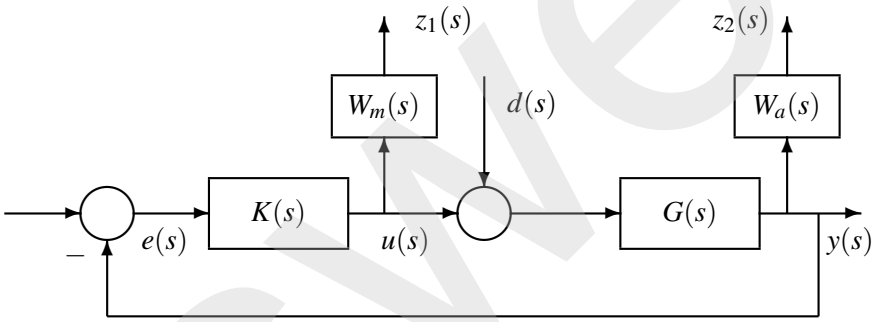
1. a) Let $\Delta(s)$ and $H(s)$ be stable. The feedback loop is stable if $\|\Delta H\|_\infty < 1$.

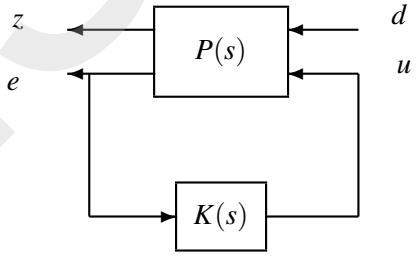
b)
- Suppose that $\Delta_a = 0$ and let the input to Δ_m be ε_2 while the output from Δ_m be δ_2. Then a calculation shows that $\varepsilon_2 = -KG(I+KG)^{-1}\delta_2$. Using the small gain theorem, to satisfy the first design requirement, it is sufficient that $\left\|\Delta_m(j\omega)K(j\omega)G(j\omega)\left(I+K(j\omega)G(j\omega)\right)^{-1}\right\| < 1, \forall\omega$. This can be satisfied if $\left\|W_m KG(I+KG)^{-1}\right\|_\infty < 1$, where $W_m = w_m I$.
- An analogous procedure shows that to satisfy the second design requirement, it is sufficient that $\left\|\Delta_a(j\omega)G(j\omega)\left(I+K(j\omega)G(j\omega)\right)^{-1}\right\| < 1, \forall\omega$. This can be satisfied if $\left\|W_a G(I+KG)^{-1}\right\|_\infty < 1$, where $W_a = w_a I$.

To satisfy the design requirements, it is sufficient that $\left\|\begin{bmatrix} W_m KG(I+KG)^{-1} \\ W_a G(I+KG)^{-1}\end{bmatrix}\right\|_\infty < 1$.

c) So, the design specifications reduce to the requirement that the transfer matrix from r to $z = [z_1^T \;\; z_2^T \;\; z_3^T]^T$ in the following diagram has $\mathscr{H}_\infty$-norm less than 1.



d) The generalized regulator problem: find an internally stabilizing K such that $\|T_{zd}\|_\infty < 1$



$$P = \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} = \left[\begin{array}{c|c} 0 & W_m \\ W_a G & W_a G \\ \hline -G & -G \end{array}\right].$$

e) If we interpret $d(s)$ as a disturbance, it follows from Part (c) that we should expect good disturbance rejection at the input u and output y.

Answers

2. a) i) The conditions are $P = P^T \succ 0$ and $A^T P + PA \prec 0$. We prove stability of A. Let $Az = \lambda z$, $z \neq 0$. Then multiplying the inequality from the left by z^T and from the right by z gives $(\lambda + \bar{\lambda}) z^T P z < 0$. Since $P \succ 0$ it follows that $z^T P z > 0$ and it follows that $\lambda + \bar{\lambda} < 0$ so that A is stable.

ii) Since A is stable, $\|H\|_\infty < \gamma$ if and only if, with $x(0) = 0$, $J := \int_0^\infty [y^T y - \gamma^2 u^T u] dt < 0$, for all $u(t)$ such that $\|u\|_2 < \infty$. If $\|u\|_2$ is bounded, then $\lim_{t\to\infty} x(t) = 0$. Now, $\int_0^\infty \frac{\mathrm{d}}{\mathrm{d}t}[x^T P x] dt = x(\infty)^T P x(\infty) - x(0)^T P x(0) = 0$. So,

$$0 = \int_0^\infty \left(\dot{x}^T P x + x^T P \dot{x}\right) dt = \int_0^\infty [x^T (A^T P + PA) x + x^T P B u + u^T B^T P x] dt.$$

Use $y = Cx + Du$ and add the last expression to J

$$\begin{aligned} J &= \int_0^\infty [x^T (A^T P + PA + C^T C) x + 2 x^T (PB + C^T D) u + u^T (D^T D - \gamma^2 I) u] dt \\ &= \int_0^\infty \begin{bmatrix} x^T & u^T \end{bmatrix} \overbrace{\begin{bmatrix} A^T P + PA + C^T C & PB + C^T D \\ B^T P + D^T C & D^T D - \gamma^2 I \end{bmatrix}}^{M} \begin{bmatrix} x \\ u \end{bmatrix} dt. \end{aligned}$$

It follows that $J < 0$, and so $\|H\|_\infty < \gamma$, if $P \succ 0$ and $M \prec 0$.

iii) Now, $H(s)^T = D^T + B^T (sI - A^T)^{-1} C^T$. It follows that the dual conditions are: there exists a $P = P^T \succ 0$ such that

$$\begin{bmatrix} AP + PA^T + BB^T & PC^T + BD^T \\ CP + DB^T & DD^T - \gamma^2 I \end{bmatrix} \prec 0.$$

b) i) The state space equations are

$$\begin{aligned} \dot{x}(t) &= Ax(t) + d + LCx(t) = (A + LC)x(t) + d \\ e(t) &= Cx(t) \end{aligned}$$

Therefore $T_{ed}(s) = C\,(sI - (A + LC))^{-1}$.

ii) Using the answer to Part a(ii), replacing A by $A + LC$ and setting $D = 0$ and $B = I$, there exists a feasible matrix L if there exists $P \succ 0$ such that

$$\begin{bmatrix} (A + LC)^T P + P(A + LC) + C^T C & P \\ P & -\gamma^2 I \end{bmatrix} \prec 0.$$

Since the only nonlinearity is due to the product PL, we let $Y = PL$ and so there exists a feasible gain matrix L if there exists $P \succ 0$ and Y such that

$$\begin{bmatrix} A^T P + PA + YC + C^T Y^T + C^T C & P \\ P & -\gamma^2 I \end{bmatrix} \prec 0.$$

Answers

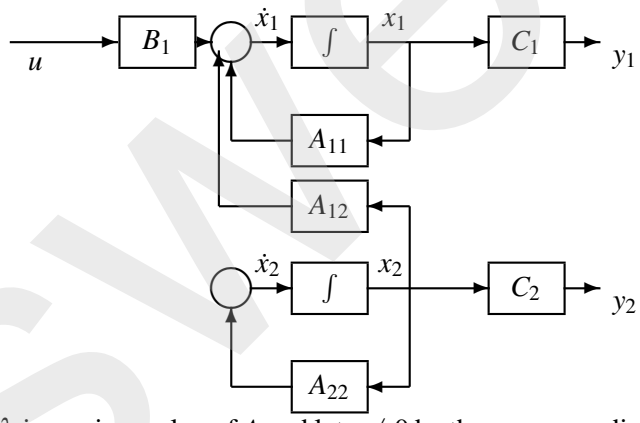
3. a) Let the realization be partitioned as $G(s) \stackrel{s}{=} \left[\begin{array}{cc|c} A_{11} & A_{12} & B_1 \\ 0 & A_{22} & 0 \\ \hline C_1 & C_2 & 0 \end{array}\right]$, where A_{11} is the leading 2×2 submatrix of A.

i) Then it follows from the zero blocks on the $(2,1)$ block entry of A and the second block entry of B that the modes corresponding to $A_{22} = 5$ are uncontrollable. Since A_{22} is unstable, the realization is not stabilizable.

ii) Removing the uncontrollable part, $G(s) \stackrel{s}{=} \left[\begin{array}{c|c} A_{11} & B_1 \\ \hline C_1 & 0 \end{array}\right]$. Since B_1 and C_1 have rank 2, it follows that this is a minimal realization. Thus the original realization of $G(s)$ is observable. By direct evaluation

$$G(s) = D + C(sI - A)^{-1}B = D + C_1(sI - A_{11})^{-1}B_1 = \left[\begin{array}{cc} \frac{1}{s+1} & 0 \\ 0 & \frac{6}{s-2} \end{array}\right].$$

iii) The minimal/nonminimal subsystems of $G(s)$ are illustrated below where $x = \left[\begin{array}{c} x_1 \\ x_2 \end{array}\right]$ and $y = y_1 + y_2$.



b) i) Suppose that λ is an eigenvalue of A and let $z \neq 0$ be the corresponding eigenvector. Then $Az = \lambda z$. Pre– and post–multiplying the matrix inequality by z' and z, respectively, we get

$$(\lambda + \bar{\lambda}) z'Qz < 0.$$

Since $z \neq 0$ and $Q \succ 0$, this implies that $z'Qz > 0$ so that $\lambda + \bar{\lambda} < 0$ and so A is stable.

ii) The condition that all the eigenvalues of A are to the left of the line $s = \sigma$ in the complex plane is equivalent to the condition that $A - \sigma I$ is stable, where I is the identity matrix. Using Part (bi) above, a sufficient condition for this is that there exists a matrix $Q = Q' \succ 0$ such that

$$(A - \sigma I)'Q + Q(A - \sigma I) \prec 0,$$

or

$$A'Q + QA - 2\sigma Q \prec 0.$$

Answers

4. a) The generalized regulator formulation is given by

$$\begin{bmatrix} z_e(s) \\ y_e(s) \end{bmatrix} = P(s) \begin{bmatrix} w(s) \\ u_e(s) \end{bmatrix},\ u_e(s) = L y_e(s), P(s) = \begin{bmatrix} P_{11}(s) & P_{12}(s) \\ P_{21}(s) & P_{22}(s) \end{bmatrix} \stackrel{s}{=} \left[\begin{array}{c|cc|c} A & B & 0 & I \\ \hline C & 0 & 0 & 0 \\ \hline C & 0 & I & 0 \end{array}\right].$$

b) The duality principles are: (1) (A,B) is controllable $\Leftrightarrow$ (A^T, B^T) is observable. (2) $G(s)$ is stable $\Leftrightarrow$ $G(s)^T$ is stable and $\|G\|_\infty = \|G^T\|_\infty$. (3) $[F_l(P,L)]^T = F_l(P^T, L^T)$. It follows that L solves the generalised regulator problem for P iff L^T solves the generalised regulator problem for P^T.

c) Taking the transpose of $P(s)$ in Part (a), redefining $A := A'$, $B := C'$, $C := B'$, $F := L^T$, $w := z_e$, $z := w$, $x := x_e$, $y := u_e$ and $u := y_e$ we get the state-feedback problem

$$\begin{bmatrix} z(s) \\ y(s) \end{bmatrix} = P(s) \begin{bmatrix} w(s) \\ u(s) \end{bmatrix},\ u(s) = F y(s), P(s) = \begin{bmatrix} P_{11}(s) & P_{12}(s) \\ P_{21}(s) & P_{22}(s) \end{bmatrix} \stackrel{s}{=} \left[\begin{array}{c|cc} A & B & B \\ \hline C & 0 & 0 \\ 0 & 0 & I \\ \hline I & 0 & 0 \end{array}\right].$$

d) The requirement $\|T_{zw}\|_\infty < \gamma$ is equivalent to $J := \|z\|_2^2 - \gamma^2 \|w\|_2^2 < 0$. Let $V = x'Xx$. Provided that $X = X' > 0$ and $\dot{V} < 0$ along the closed-loop trajectory, we can assume $\lim_{t\to\infty} x(t) = 0$. Then

$$\dot{V} = \dot{x}'Xx + x'X\dot{x} = x'\left(A'X + XA + F'BX + XB'F\right)x + x'XBw + w'B'Xx.$$

Integrating from 0 to ∞ and using $x(0) = x(\infty) = 0$,

$$0 = \int_0^\infty [x'\left(A'X + XA + F'B'X + XBF\right)x + x'XBw + w'B'Xx]dt.$$

Using the definition of J and adding the last equation, $J =$

$$\int_0^\infty \{x'[A'X + XA + C'C + F'F + F'B'X + XBF]x - [\gamma^{-2} w'w - x'XBw - w'B'Xx]\}dt.$$

Completing the squares by using

$$\begin{aligned} (F + B'X)'(F + B'X) &= F'F + F'B'X + XBF + XBB'X \\ \|(\gamma w - \gamma^{-1} B'Xx)\|^2 &= \gamma^2 w'w - w'B'Xx - x'XBw + \gamma^{-2} x'XBB'Xx, \end{aligned}$$

$$J = \int_0^\infty \{x'[A'X + XA + C'C - (1-\gamma^{-2})XBB'X]x + \|(F + B'X)x\|^2 - \left\|\gamma w - \gamma^{-1} B'Xx\right\|^2\}dt.$$

e) Thus two sufficient conditions for $J < 0$ are the existence of X such that

$$A'X + XA + C'C - (1-\gamma^{-2})XBB'X = 0, \qquad X = X' > 0.$$

The feedback gain is obtained as $F = -B'X$. The worst case disturbance is $w^* = \gamma^{-2} B'Xx$. The closed-loop with $u = Fx$ and $w = w^*$ is $\dot{x} = [A - (1-\gamma^{-2})BB'X]x$ and a third condition is $Re\ \lambda_i[A - (1-\gamma^{-2})BB'X] < 0,\ \forall\ i$. It remains to prove $\dot{V} < 0$ for $u = Fx$ and $w = 0$. But

$$\dot{V} = -x'(C'C + (1+\gamma^{-2})XBB'X)x < 0$$

for all $x \neq 0$ proving closed-loop stability since (A,C) is assumed observable.

Answers